Evolutionary history from genomes to single cells by lossless compression

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One-Sentence Summary: Lossless data compression unifies evolutionary trees, rapid radiations, gene transfer, and convergence across scales of life.

Abstract

Reconstructing evolutionary history is central to biology and medicine, yet standard pipelines rely on separate models and arbitrary cutoffs to build trees, collapse weakly supported branches, and detect recombination or convergent evolution. Because inheritance copies an ancestor with sparse edits, we recast evolutionary tree and network inference as lossless data compression. Implemented in Bushwhacker, a branch or lateral link is added only when the shared changes it explains save more information than the connection costs to describe. Rapid bursts without shared mutations remain naturally unresolved as multi-way hubs, transferred sequence tracts separate from convergent sites, and Tree-of-Life phylogenies run one to two orders of magnitude faster than maximum likelihood in a fraction of the memory. Across immune and tumor cells, pandemic viruses, microbial domains, and mammalian coding genes and proteomes, compression resolves apparent gene-tree conflicts, predicts clonal selection, maps recombinants, and separates paralog gene conversion from adaptive convergence.

Evolutionary trees are among the most widely used structures in science: they organize Earth's biodiversity, track unfolding epidemics, trace immune and tumor clones inside individual patients, and anchor comparative genomics across the Tree of Life. Yet even as sequencing has expanded from single genes to millions of genomes, the conceptual foundations used to reconstruct trees have changed remarkably little for nearly half a century. Since distance clustering, parsimony, and probabilistic Markovian likelihood were introduced between the 1960s and early 1980s (*1*–*3*), major advances have largely focused on scaling tree search and enriching parametric substitution models within those classical paradigms (*4*–*6*). Genuinely new mathematical perspectives on how evolutionary history can be inferred have been rare.

Here we introduce a different foundation, rooted in information theory: lossless data compression (*7*, *8*). Biological inheritance is, at heart, a copying process with sparse edits. Whether

an ancestor transmits a genome, a protein, or a somatic cellular profile, related lineages share most of their states because they inherited them from a common source. Recording shared changes once along an ancestral branch is therefore far more economical than recording them separately in every descendant—making an evolutionary graph the natural coordinate system for compressing biological data (Fig. 1). This equivalence is not merely metaphorical: encoding viral and microbial cohorts along an inferred phylogeny physically compresses genomes substantially further than general-purpose algorithms, until pervasive recombination breaks tree-like descent itself.

Under Rissanen's Minimum Description Length (MDL) principle, both the evolutionary graph and the changes along it are measured in the same unit of information (natural logarithmic units, or nats; 1 nat $\approx$ 1.44 bits). Every proposed connection must earn its place by shortening the total description of the data: a branch is added only when the shared changes it explains save more information than the branch itself costs to specify. Because every prefix-free lossless code length L defines a code probability $P \propto e^{-L}$ (*7*, *8*), compression places trees, models, and networks on a unified probabilistic scale: given a graph inferred by discrete heuristic tree search, comparing nested structures with and without an individual branch or lateral link yields its local log-odds evidence ΔL and conditional two-hypothesis posterior support $P_{\text{local}} = 1/(1 + e^{-\Delta L})$. That local criterion replaces separate pruning thresholds and bootstrap replicates (*9*–*11*). When lineages diversify faster than mutations accumulate—during cellular proliferation, epidemic super-spreading, or rapid species radiations—newly diverging lineages share no mutations that justify grouping them into sub-branches. Adding an internal branch then requires more information to specify which lineages it groups than it saves in shared edits, leaving rapid bursts naturally unresolved as multi-way hubs (polytomies) rather than forcing strictly bifurcating branches that fluctuate across models or create apparent conflict among genes (*12*–*14*). When lineages diverge unevenly or across billion-year timescales, coincidental matches along mutation-dense branches or rapidly mutating sites save far less positional description length than shared changes on nearly invariant stems, avoiding the long-branch artifacts of unweighted mutation counting (*15*) and connecting discrete edit coding seamlessly to continuous substitution probabilities (*16*–*18*). And once the vertical tree is established, lateral links between distant lineages can be evaluated in the same currency, separating contiguous transferred tracts—from recombination, horizontal gene transfer, or nonreciprocal copying between related gene copies (paralog gene conversion) (*19*–*21*)—from scattered sites of adaptive convergence (*22*, *23*).

Implemented in our open-source engine Bushwhacker, this formulation unlocks four capabilities beyond existing phylogenetic pipelines. First, it reconstructs Tree-of-Life DNA and protein supermatrices 24- to 54-fold faster than fast maximum-likelihood search (142- to 288-fold faster than approximate maximum likelihood) in 10- to 40-fold less memory, delivering a probabilistically annotated graph—not a bare point-estimate tree—in one deterministic run without model-selection sweeps or bootstrap resampling. Second, it naturally preserves rapid evolutionary bursts as multifurcations instead of inventing arbitrary strictly bifurcating splits that flip across substitution models, search perturbations, and loci (*14*, *24*). Third, it jointly infers vertical trees and lateral networks on one ledger, separating contiguous recombination, horizontal gene transfer, and paralog gene conversion from scattered sites of adaptive convergence that otherwise distort clades (*25*–*27*). Fourth, it generalizes natively beyond nucleotide and protein alignments to discrete single-cell somatic mutations and ordered copy-number states while compiling genome cohorts into lossless `.bwz` archives physically smaller than general-purpose compressors. Across eukaryotic radiations, 882 mammalian coding exomes and proteomes (*28*), pandemic viruses, prokaryotic domains (*29*, *30*), and single-cell immune and tumor cohorts (*31*, *32*), lossless compression resolves apparent gene-tree conflict, separates paralog gene conversion from ecological convergence, maps mosaic recombinants, and predicts in vivo clonal selection directly from sequence or somatic mutation profiles.

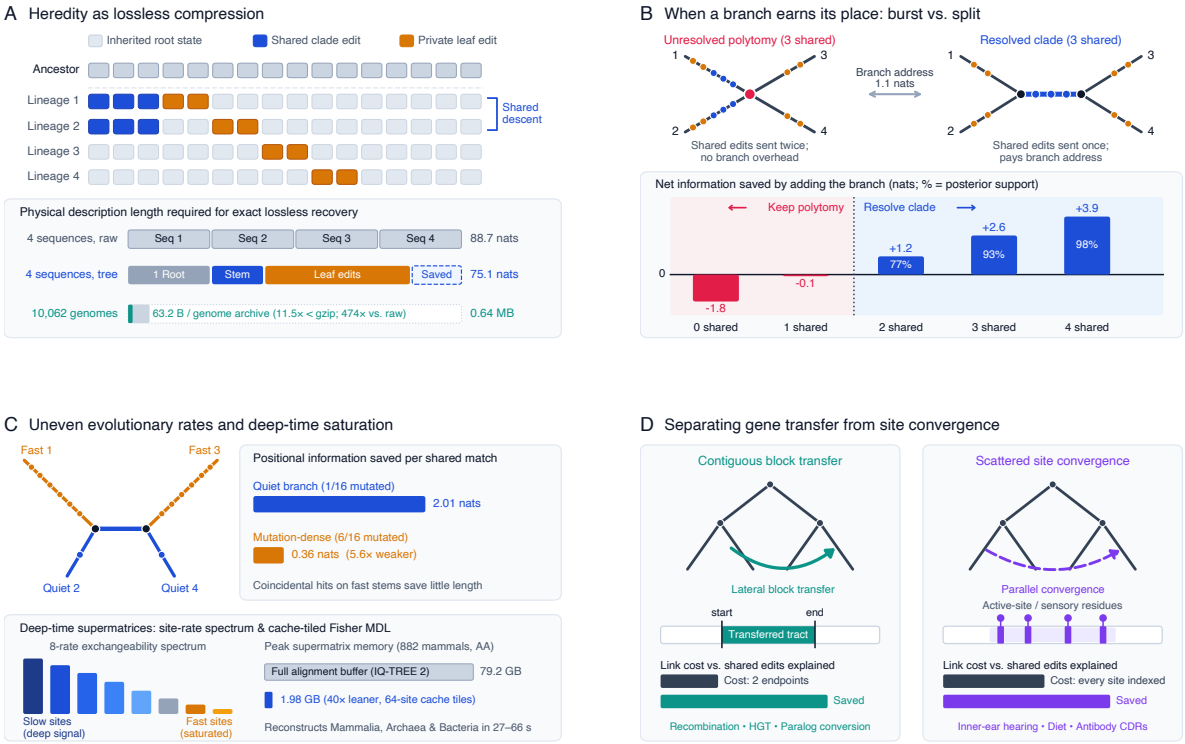


Figure 1: **Evolutionary trees and networks from lossless data compression.** (**A**) Rather than transmitting full sequences independently (88.7 nats across four 16-site sequences; 1 nat ≈ 1.44 bits), encoding one ancestral root plus sparse branch edits shortens the lossless description (75.1 nats), compounding to 63.2 bytes per 29.9-kb genome archive (11.5× smaller than `gzip -9`; 474× versus raw; Table S9) across 10,062 SARS-CoV-2 genomes. (**B**) Resolving a multi-way burst (polytomy) into a new clade transmits shared edits once rather than twice (illustrated top for three shared edits), but must pay the topological address cost of naming the split (1.1 nats for a quartet). Across zero to four shared changes (bottom), zero or one shared mutation incurs a net loss (−1.8 and −0.1 nats; conditional local posterior $P_{\text{local}} = 14\%$ and 47%) and remains a natural polytomy, whereas two to four shared changes yield positive net savings (+1.2 to +3.9 nats; $P_{\text{local}} = 77\%$ to 98%). (**C**) Positional coding automatically saves 5.6× more information per shared match on quiet stems than on mutation-dense branches, while eight-category rate-model coding in 64-site CPU cache tiles extends the same ledger to billion-year supermatrices in under 2.8 GB of memory. (**D**) On the established tree, contiguous lateral transfers pay only two endpoint coordinates to explain many linked changes, whereas scattered parallel sites must pay each site's coordinate individually, filtering out coincidental parallel mutations (homoplasy).

Building evolutionary trees and networks by lossless compression

To evaluate how concisely an evolutionary graph explains a set of biological sequences or cellular character profiles, Bushwhacker encodes two parts without loss: the graph itself and the state changes along its edges (Fig. 1A, Fig. S1, Table S5, and Materials and Methods). Starting from one ancestral sequence at the root, each branch records only how many positions changed, where those changes occurred, and what new nucleotide, amino-acid, or somatic states appeared, weighted by empirical exchange probabilities (*18*, *33*). Resolving a multifurcating burst into a new ancestral branch avoids repeating shared changes independently in each descendant, but must pay the address cost of specifying which lineages join the clade and encoding the new branch itself (Fig. 1B). Because code lengths in nats correspond directly to negative log-

probabilities, comparing nested topologies with and without a candidate branch or lateral link gives its log-odds evidence ΔL and conditional local posterior support $P_{\text{local}} = 1/(1 + e^{-\Delta L})$ conditioned on the remainder of the graph, eliminating the need to evaluate branch support with bootstrap replicates. On a four-taxon burst, zero or one shared mutation costs more to specify than it saves ($\Delta L = -1.8$ and -0.1 nats; $P_{\text{local}} = 14\%$ and 47%), leaving the burst unresolved, whereas two to four shared changes yield positive evidence ($+1.2$ to $+3.9$ nats; $P_{\text{local}} = 77\%$ to 98%).

Two further consequences follow from pricing positions and connections in the same currency (Fig. 1C,D and Supplementary Text). First, specifying an edit's location costs far more on a nearly invariant ("quiet") branch than on a mutation-dense one; consequently, moving a shared change onto an ancestral stem from quiet branches saves substantially more information than grouping coincidental matches on fast-evolving branches (5.6-fold in Fig. 1C), recovering likelihood-like rate weighting (*2*) from combinatorics alone. When divergence spans hundreds of millions or billions of years and individual sites mutate repeatedly, this discrete code generalizes smoothly within the same two-part ledger to continuous-time eight-category substitution rate models (*8*, *16*–*18*), evaluated in compact CPU cache tiles that avoid gigabyte-scale memory tables (Materials and Methods). Second, once the vertical tree is established by topological refinement, candidate lateral links between distant lineages are priced by their spatial footprint: a contiguous transferred tract pays only two genomic endpoint coordinates to explain many linked changes, whereas scattered parallel replacements must pay to address each site individually across the variable-column pool, separating recombination, horizontal gene transfer, and paralog gene conversion from site-specific adaptive convergence.

Fast, accurate trees—and limits of strictly bifurcating models

Across Tree-of-Life supermatrices, Bushwhacker incurs a modest topological accuracy cost (0.5 to 2.5 percentage points in normalized Robinson–Foulds split distance: nRF = 0.081–0.247 versus 0.076–0.236 for `IQ-TREE 2` and 0.074–0.228 including `FastTreeMP`, while halving false splits on mammalian proteins, 32 versus 74–76, and sacrificing recall mainly on short invariant stems; Table S11) while matching higher-rank taxonomic monophyly of fast maximum-likelihood tools (`IQ-TREE 2` (*6*) and `FastTreeMP` (*4*)) at a small fraction of their time and memory (Fig. 2A and Tables S2, S3, and S11). Across long-branch simulations (*15*, *18*, *34*) and genome-scale alignments of mammals (*28*), archaea, and bacteria (*30*), it reconstructs thousand-species trees in 24 to 66 s on 32 cores—24- to 54-fold faster than `IQ-TREE 2` and 142- to 288-fold faster than `FastTreeMP`—using 1.98 to 2.71 GB of memory rather than 26 to 79 GB (10- to 40-fold less). Whereas unweighted parsimony collapses on billion-year prokaryotic divergences, rate-weighted compression recovers mammalian orders (100%) and bacterial phyla (94.3%) alongside likelihood baselines. Across $44,337$ tested branches, conditional local branch posteriors are well-calibrated in the high-confidence regime ($P_{\text{local}} \geq 99\%$: 89.4% of retained splits, 98.3% true; overall expected calibration error ECE = 0.0373), while remaining optimistic across the 10.6% of intermediate splits ($P_{\text{local}} \in [0.50, 0.99)$: 40.6%–80.3% true; Table S3) because P_{local} conditions on a single optimized surrounding topology rather than marginalizing competing trees (Fig. 2B).

Standard tree builders face systematic limits, however, when evolutionary relationships depart from a single strictly bifurcating tree (Fig. 2C,D, Figs. S2 and S3, and Tables S2 to S5). Across $37,020$ ground-truth simulations spanning nucleotides, proteins, coalescent histories, lateral networks, and single cells, we evaluated three common regimes. First, when lineages diversify faster than mutations arise, likelihood methods fit chance similarities and invent 23 to 61 false splits per burst—even when statistical polytomy tests are enabled (*14*, *35*)—whereas Bushwhacker leaves mutation-free bursts unresolved (0.00 to 0.18 false splits per tree, maintaining $> 94\%$ split precision) and improves downstream coalescent species-tree reconstruction (*36*, *37*). Second, when distant lineages exchange DNA tracts or evolve parallel adaptive changes, purely tree-shaped models pull unrelated clades together or invoke excess convergent

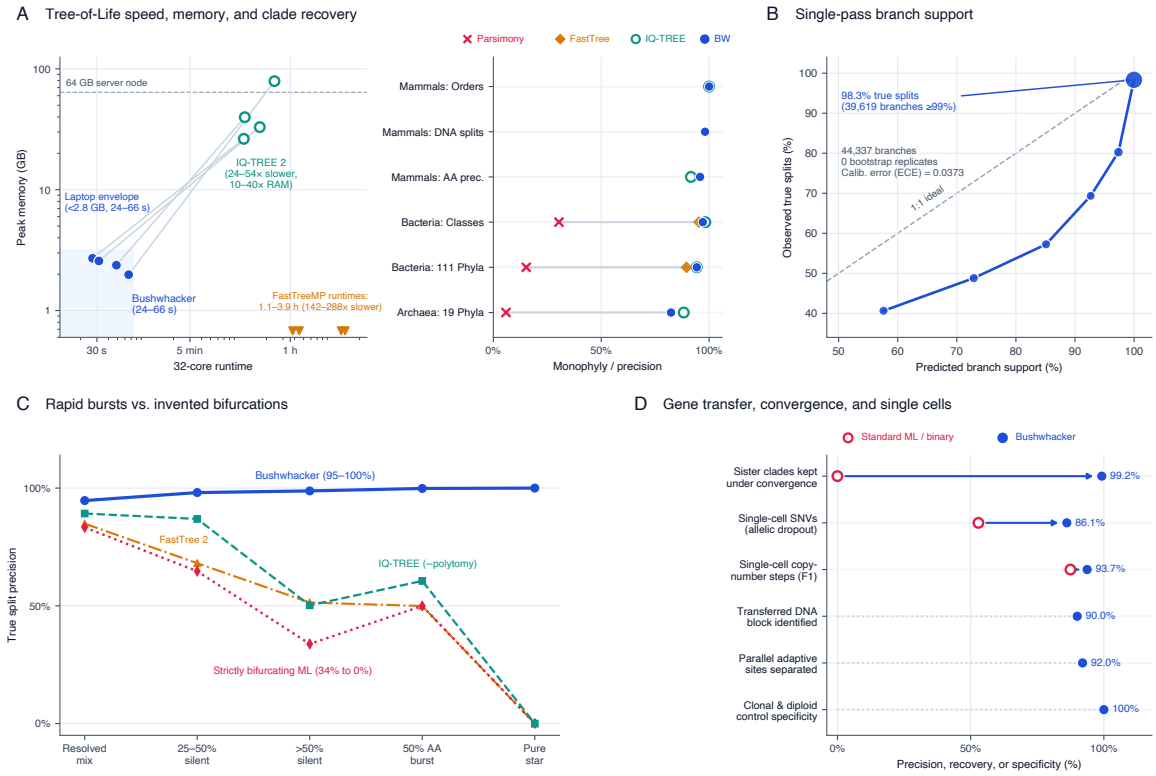


Figure 2: **Tree-of-Life speed, laptop-scale memory, bootstrap-free support, and resolution of rapid-burst and non-tree regimes.** (**A**) Thirty-two-core runtime and peak memory across four Tree-of-Life supermatrices (882 mammals, 543, 864 bp DNA and 181, 288 amino acids; 1, 672 archaea; 1, 000 bacteria; left, downward triangles mark `FastTreeMP` runtimes) and taxonomic monophyly or split precision against `IQ-TREE 2`, `FastTreeMP`, and an unweighted parsimony ablation (right; Table S11). (**B**) Bootstrap-free conditional branch support versus observed split accuracy across 44, 337 branches without bootstrap replicates (expected calibration error ECE = 0.0373; circle area proportional to branch count; Table S3). (**C**) Split precision across 36, 120 simulated nucleotide, protein, and coalescent histories (Table S2). (**D**) Split precision, event recovery, and control specificity across 800 benchmarks of horizontal DNA transfer, parallel adaptive convergence, and error-prone single-cell somatic states (400 clonal and diploid controls).

changes; pricing lateral links on the same ledger separates transferred tracts from convergent sites (95% to 100% recovery) with zero false positives on tree-like controls, keeping the vertical species tree intact. Third, on error-prone single-cell mutation and copy-number profiles subject to allelic dropout, native multi-state and ordered-step compression improves clade precision and recovery over binary maximum likelihood and binarized coding while calling zero false clones on healthy diploid controls.

Mutation-free branches explain most apparent gene-tree conflict

Roughly two-thirds of published single-gene trees clash with genome-wide phylogenies, fueling debates over hybridization and lineage sorting (*12*, *13*). However, a major share of these clashes arises without biological discordance: merely changing the substitution model or starting search perturbation routinely shifts the tree inferred from the exact same alignment (*14*, *24*, *35*). Both problems stem from enforcing a strict bifurcation rule onto mutation-free lineages. When speciation outpaces mutation in an individual locus, brief ancestral stems record zero

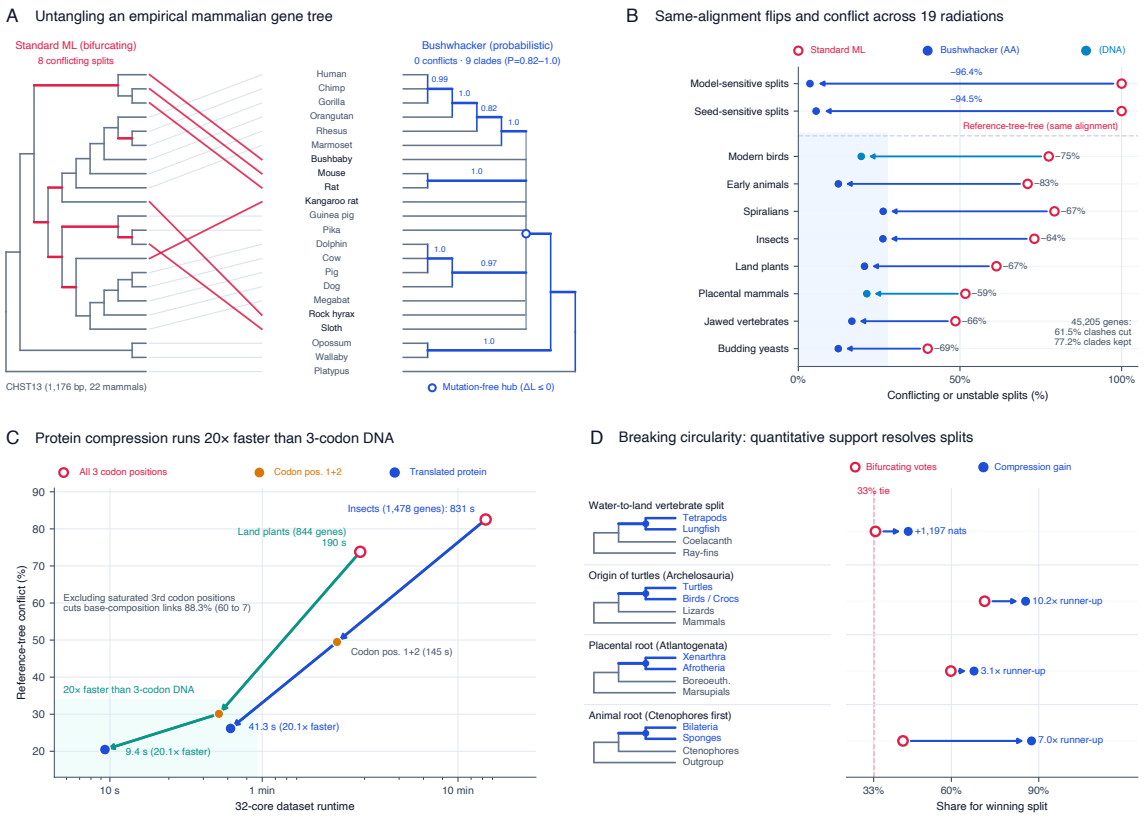


Figure 3: **Mutation-free branches explain apparent gene-tree conflict against species phylogenies, same-alignment instability, and contested splits across eukaryotic radiations.** (**A**) Empirical tanglegram for mammalian *CHST13* (1,176 bp, 22 species), contrasting the published strictly bifurcating maximum-likelihood tree (8 conflicting splits in crimson; crossed ribbons highlight displaced taxa) against Bushwhacker's reference-free probabilistic tree (0 conflicts; 9 clades retained with conditional branch posteriors $P_{\text{local}} = 0.82$–1.0; open circle marks the $\Delta L \leq 0$ placental hub). (**B**) Reference-tree-free controls showing 96.4% removal of model-sensitive splits (7,200 genes) and 94.5% removal of seed-sensitive splits across 20 independent RAxML/IQ-TREE searches per locus (83,180 runs across 4,159 alignments; Table S6) on identical alignments (top), above conflict reduction against published genome-wide reference trees across 45,205 genes in 19 radiations (61.5% of clashes removed; 77.2% of shared clades retained; dark blue, amino-acid datasets; sky blue, DNA datasets; bottom). (**C**) Dataset runtime and reference-tree conflict from all three codon positions (strictly bifurcating) to codon positions 1+2 and translated 20-state proteins across land plants (844 genes) and insects (1,478 genes), running 20× faster than three-codon DNA while excluding saturated third-codon positions cuts plant base-composition links 88.3% (60 to 7). (**D**) Synchronized four-taxon topologies and winning-split shares across three competing hypotheses before (strictly bifurcating votes) and after (cumulative compression gain) withholding mutation-free splits to break species-tree circularity (dashed line, 33% three-way tie).

informative changes; leaving these mutation-free branches unresolved ($\Delta L \leq 0$, $P_{\text{local}} \leq 50\%$) states probabilistically that the gene cannot distinguish competing branching orders from a coin flip. On a typical mammalian gene (*CHST13*; Fig. 3A), enforced bifurcations in maximum likelihood create eight conflicting splits—pulling bushbabies onto rodents and rock hyraxes into primates—whereas Bushwhacker retains nine supported clades and leaves the mutation-free placental hub open with zero conflicts.

Across 45,205 genes in 19 eukaryotic radiations, withholding zero-evidence splits removes estimation noise rather than biological signal (Fig. 3B,C and Tables S1 and S6). With no reference tree involved, 95% to 96% of branches that flip across substitution models or independent RAxML/IQ-TREE searches on identical alignments carry $\Delta L \leq 0$ and disappear in one deterministic run. Measured against published genome-wide phylogenies, withholding $\Delta L \leq 0$ splits eliminates 61.5% of apparent conflicts (74.9% in early birds (*38*, *39*) and 82.5% in early animals (*12*)) while retaining 77.2% of shared clades—nearly invariant to how the reference phylogeny itself was built (*40*)—and compressing translated protein sequences rather than coding DNA where third codon positions are mutationally saturated strips away base-composition biases in deep plant and insect radiations (*41*).

Crucially, because species trees are themselves inferred from genes, forcing mutation-free loci into strictly bifurcating trees creates a circular trap: thousands of uninformative genes cast random coin-flip votes that destabilize contested deep nodes (*13*) (Fig. 3D). Weighting branches by their positive compression support ($\Delta L > 0$) breaks that circularity—resolving lungfish as sister to tetrapods where strictly bifurcating gene counts stall in a three-way tie (*42*), uniting turtles with archosaurs > 10-fold, placing ctenophores as the earliest animal lineage (*43*, *44*), and favoring Atlantogenata at the placental root (*45*). Passing multifurcating Bushwhacker gene trees directly to `ASTRAL-IV` (*37*) likewise lowers coalescent species-tree error without bootstrap cutoffs across rapid-speciation "anomaly-zone" simulations—where the most frequent strictly bifurcating gene tree disagrees with the true species tree (Table S2)—and 11,164 distinct empirical loci (12,008 alignments) spanning animals, plants, fungi, birds (*46*), and mammals (Table S1).

Lineage sorting, paralog gene conversion, and convergence across mammalian proteomes

Measuring incomplete lineage sorting, paralog gene conversion, and adaptive convergence together across whole proteomes has been out of reach for likelihood pipelines, which cannot scale to thousands of hundreds-of-species alignments and routinely let unaccounted gene conversion or convergent sites distort single-gene and protein phylogenies. On the outer hair-cell motor gene *Prestin* (*SLC26A5*; 116 mammals), where sonar-linked replacements famously pull echolocating dolphins inside bats on standard protein trees (*22*, *23*), Bushwhacker keeps whales inside Cetartiodactyla ($P_{\text{local}} = 1.00$) and bats monophyletic ($P_{\text{local}} = 1.00$; 93.8% species-split precision) while isolating a lateral convergence link (+3.9 nats, $P_{\text{local}} = 98.0\%$) between dolphins and horseshoe bats across four shared motor-domain codons (Fig. 4A and Fig. S4). Scaling that unified ledger to 19,952 orthologous coding genes (19,953 proteins) across up to 882 mammals (*28*, *47*) shows how strongly a single organismal tree organizes the mammalian proteome (Fig. 4B, Fig. S5, and Tables S1 and S7): 85.9% of mammalian proteins—rising to 97.9% above 1,500 amino acids—compress more tightly on the mammalian species phylogeny than on an unconstrained gene tree, while the fraction of candidate species-tree branches showing genuine incomplete lineage sorting in proteins (19.8% overall) climbs with protein length toward the 20.8% ancestral polymorphism baseline measured in coding DNA.

Without using synteny or paralog labels, spatial pricing separates two lateral processes across those proteins (Fig. 4C). Contiguous paralog gene conversion is detected in 6.0% of mammalian proteins (median tract 85 residues), acts largely within families (68.9%; 88.3% within orders), and enriches 3.0- to 6.6-fold inside tandemly arrayed clusters—including globins, detoxification enzymes, olfactory receptors, zinc-finger arrays, and cadherins or keratins—versus 4.2% in other coding loci. Scattered parallel amino-acid changes, by contrast, are inferred in 11.9% of proteins and cross mammalian orders 38.0% of the time (3.3-fold higher cross-order share than paralog gene conversion), identifying candidate sites under convergent selection.

Those cross-order parallel sites reveal that ecological adaptation is sharply focal rather than proteome-wide (Fig. 4D and Fig. S5C). Resolving whether sonar drove convergence across thou-

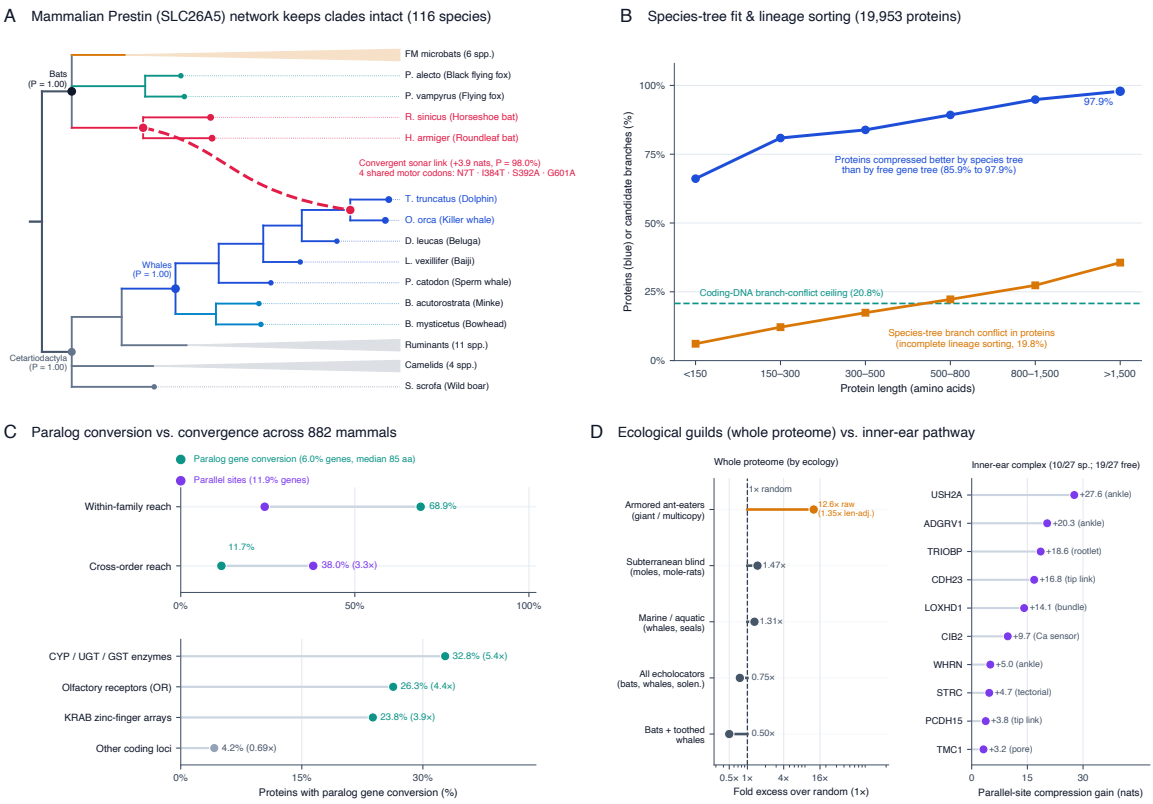


Figure 4: **Proteome-wide census of organismal inheritance, paralog gene conversion, and adaptive convergence across 19,953 proteins in 882 mammals.** (**A**) Empirical Bushwhacker network for mammalian *Prestin* (*SLC26A5*, 116 species, 2,235 bp; Laurasiatherian clades shown), preserving monophyletic bats ($P_{\text{local}} = 1.00$) and whales inside Cetartiodactyla ($P_{\text{local}} = 1.00$) while isolating the lateral convergence link (+3.9 nats, $P_{\text{local}} = 98.0\%$) between echolocating horseshoe/roundleaf bats and dolphins across four shared motor-domain codons (`N7T`, `I384T`, `S392A`, `G601A`). (**B**) Percentage of proteins compressing more tightly on the mammalian species tree than on a free gene tree (85.9% overall; 97.9% above 1,500 residues, blue) alongside the percentage of candidate species-tree branches showing genuine protein incomplete lineage sorting ($\Delta L > 0$: 19.8% overall, amber, approaching the 20.8% coding-DNA conflict ceiling, teal; Table S7). (**C**) Spatial and taxonomic separation of 8,670 lateral conversion or convergence links inferred on the species tree without synteny or paralog labels: contiguous paralog gene conversion (6.0% of proteins, $n = 2,251$ tracts, median 85 residues) acts locally (68.9% within families, 88.3% within orders) and enriches 3.0–6.6× in tandemly arrayed gene families versus 4.2% (0.69×) in other coding loci, whereas scattered parallel-site links (11.9% of proteins, $n = 6,419$) cross mammalian orders 38.0% of the time (3.3×). (**D**) Proteome-wide cross-order parallel-site enrichment by ecological guild relative to random expectation (1×; left)—showing neutrality across echolocators (0.50–0.75×), marine mammals (1.31×), and subterranean blind mammals (1.47×), while the raw 12.6× excess in armored ant- and termite-eating mammals (4 links) traces to giant proteins (1.35× length-adjusted) and multicopy families—contrasted with focal parallel-site compression gains (nats) across 10/27 fixed species-tree proteins (19/27 on free gene trees) in the inner-ear cochlear mechanotransduction complex (right; Table S7).

sands of proteins (*25*) or reflects background noise (*26*, *48*), echolocating bats, toothed whales, and solenodons show no proteome-wide excess (0.50–0.75× random expectation), and marine (1.31×) and subterranean blind (1.47×) mammals are likewise neutral across the proteome.

Even the raw 12.6× excess in armored ant- and termite-eating pangolins, anteaters, and armadillos (four links) traces largely to giant muscle and centrosomal proteins (*TTN*, *ALMS1*; 1.35× after length adjustment) and multicopy receptor or zinc-finger families (*OR10D3*, *ZSCAN16*; Table S7). Instead, adaptive convergence concentrates in shared physiological pathways: positive parallel-site compression links appear across 70% (19 of 27 genes; 10 of 27 on the fixed species tree, plus 116-species *Prestin*) of the inner-ear hair-cell mechanotransduction complex—spanning stereocilia tip and ankle links (*USH2A*, *ADGRV1*, *CDH23*, *PCDH15*), the *TRIOBP* rootlet bundler, the *LOXHD1* and *STRC* connectors, and the *TMC1–CIB2* channel complex—much as C_4 photosynthesis concentrates parallel changes in catalytic grass enzymes (*49*).

Outbreak hubs, mosaic viruses, and microbial gene transfer

Pathogen evolution is characterized by processes that are poorly reflected by strictly bifurcating trees: super-spreading transmits identical or single-mutation genomes to dozens of hosts before substitutions accumulate, co-infections facilitate frequent recombination between divergent strains, and selection for immune escape or drug resistance drives parallel adaptive mutations across multitudes of independent lineages (*20*, *27*). Whereas strictly bifurcating trees invent arbitrary transmission chains inside acute outbreaks that are better represented as polytomies, and likelihood recombination scans cannot scale to pandemic cohorts (*11*), Bushwhacker resolves outbreak hubs, recombinational transfers, and convergent mutations in a single deterministic run (Fig. 5A,C and Tables S1 and S7). Across 10,062 SARS-CoV-2 and 5,498 dengue virus genomes, compression preserves 1,426 SARS-CoV-2 super-spreading hubs of up to 335 descendants alongside urban dengue bursts—which nearly double on translated proteins where silent nucleotide changes no longer separate rapid transmissions—while mapping SARS-CoV-2 Spike and Omicron mosaic tracts, natural dengue recombinants (*20*), and HIV-1 recombination breakpoints (*50*).

Encoding genome cohorts into lossless `.bwz` archives via arithmetic coding (*51*) also tests directly whether shared descent is the main physical regularity of biological sequences (Fig. 5B and Table S9). Storing 10,062 SARS-CoV-2 genomes requires 300.9 MB in uncompressed form and only 636 KB as a complete lossless archive (63.2 bytes per viral genome; 1.25× smaller than `xz -6` and 11.5× smaller than `gzip -9`), or 388 KB for the compressed sequence payload alone (up to 2.06× smaller than `xz -6` and 18.9× smaller than `gzip -9`), with consistent gains across influenza, dengue, mitochondrial, plastid, and whole bacterial chromosomes. Equally telling is when that advantage reverses: across controlled recombination titrations ($K = 0 \to 32$), mosaic bat sarbecoviruses (0.71×) (*52*, *53*), and recombinant HIV-1 (0.65×), pervasive whole-genome mosaicism erases any single tree and lets multi-donor string dictionaries overtake single-tree coding, making archive size a direct gauge of tree-like versus mosaic evolution.

In prokaryotes, pricing vertical splits and horizontal transfers in the same unit turns qualitative debates over a microbial "Tree versus Web of Life" (*19*, *54*) and the Complexity Hypothesis (*29*, *55*) into a quantitative budget (Fig. 5D and Table S8). In hospital outbreaks of carbapenem-resistant *Klebsiella pneumoniae* (*56*), the 214-kb core chromosome resolves every branch across nine clinical isolates (6/6 splits at $P_{\text{local}} = 1.00$, with lateral transfer contributing just 0.5% of the chromosomal budget), whereas the 882-bp *blaKPC* plasmid carried by those exact same strains collapses 5 of 6 splits into multifurcating *blaKPC-2* and *blaKPC-3* star hubs separated by a single nucleotide (`C814T`) as horizontal plasmid spread outpaces mutation. Across 122 marker proteins in 1,672 archaeal species (*30*), horizontal transfers likewise appear exclusively in operational metabolic and regulatory genes (11.8%, 12/102—including cross-phylum highways between symbiotic DPANN archaea and their hosts (*57*)) versus 0% (0/20, two-sided Fisher $p = 0.21$) in core ribosomal and RNA polymerase subunits; and as single-gene microbiome 16S rRNA trees scale from 256 to 32,768 taxa (*58*), supported splits saturate smoothly from 91.4% ($N = 1,024$) to 76.0% ($N = 32,768$), cutting taxonomy-contradicting splits per

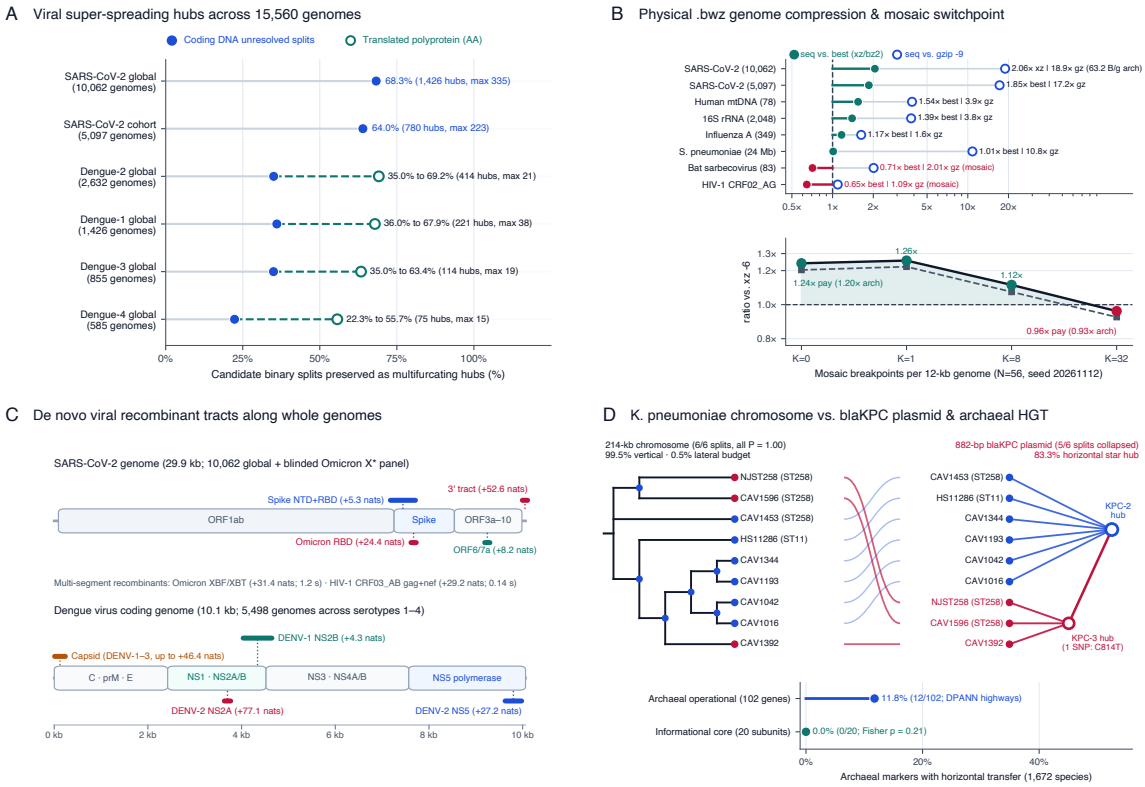


Figure 5: **Viral super-spreading hubs, physical genome compression, whole-genome recombinant mapping, and microbial gene-transfer budgets.** (**A**) Candidate strictly bifurcating splits preserved as multifurcating outbreak hubs and largest hub sizes across 10,062 global SARS-CoV-2 genomes (68.3% unresolved; 1,426 hubs up to 335 descendants) and 5,498 global dengue genomes across serotypes 1–4 in coding DNA versus translated polyproteins. (**B**) Physical lossless `.bwz` sequence payload and full archive size reductions over `xz -6`/`bzip2 -9` (up to 2.06× sequence payload) and `gzip -9` (up to 18.9×; 38.5 B/genome payload and 63.2 B/genome total archive on SARS-CoV-2; top), contrasted with the crossover below 1× under whole-genome mosaicism across controlled breakpoint titrations ($K = 0 \rightarrow 32$; bottom), bat sarbecoviruses (0.71×), and recombinant HIV-1 (0.65×). (**C**) Inferred mosaic recombinant tracts mapped along the 29.9-kb SARS-CoV-2 and 10.1-kb dengue genomes in a single run. (**D**) Side-by-side tanglegram of nine carbapenem-resistant clinical *Klebsiella pneumoniae* isolates on the 214-kb core chromosome (6/6 splits resolved at $P_{\text{local}} = 1.00$; 99.5% vertical budget) versus their 882-bp *blaKPC* carbapenemase plasmid (5/6 splits collapsed into horizontal *blaKPC-2* and *blaKPC-3* [`C814T`] star hubs; top), above horizontal transfer rates across 122 archaeal marker proteins in 1,672 species (11.8% operational versus 0.0% informational core subunits, two-sided Fisher $p = 0.21$; bottom).

group by 19% to 24% while running 4.5× faster than `FastTreeMP` (Table S1).

Somatic selection and clonal evolution in single cells

Inside germinal centers and tumors, positively selected cells divide so rapidly that a single founder spawns a star-like burst of daughter lineages before new somatic mutations arise (*59*). Whereas specialized lineage tools recover these bursts only when given cell-abundance counts or hypermutation models (*60*), reconstructing 147 mouse germinal centers (10, 429 B cells across 29 mice, 9, 352 unique genotypes (*31*)) from deduplicated sequences alone shows that compression geometry reads out Darwinian selection directly (Fig. 6A,B and Table S1).

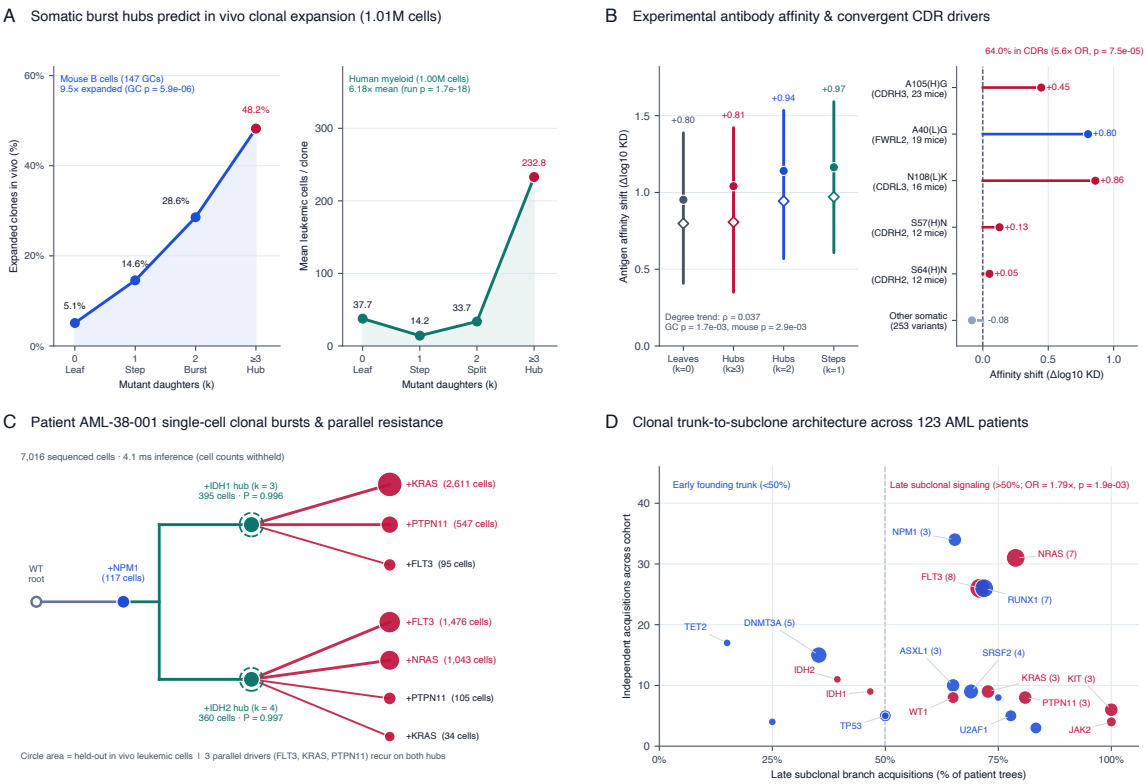


Figure 6: **Somatic compression hubs predict in vivo clonal selection and parallel resistance across single-cell immune repertoires and human leukemia.** (**A**) With all cell counts withheld during inference, inferred mutant daughter degree k monotonically predicts in vivo clonal expansion across 147 mouse B-cell germinal centers (10,429 cells (*31*); 5.1% $\rightarrow$ 48.2%, 9.5×, hierarchical GC paired Wilcoxon $p = 5.9 \times 10^{-6}$; left) and 168 human myeloid single-cell DNA cohorts (1,001,609 cells; 37.7 $\rightarrow$ 232.8 mean cells per clone, 6.18×, per-run paired Wilcoxon $p = 1.7 \times 10^{-18}$; right). (**B**) Independent yeast-display antigen-binding affinity ($\Delta \log_{10} K_D$) across inferred B-cell node classes (hierarchical within-GC trend $p = 1.7 \times 10^{-3}$, within-mouse $p = 2.9 \times 10^{-3}$; left) and across parallel cross-mouse mutations (64.0% inside complementarity-determining regions versus 24.1% background, 5.6× odds ratio, $p = 7.5 \times 10^{-5}$; right). (**C**) Empirical single-cell leukemic clonal phylogeny of human patient `AML-38-001` (7,016 sequenced cells), reconstructed in 4.1 ms with cell counts withheld (circle area proportional to held-out in vivo cell count): an early *NPM1* founder splits into parallel *IDH1* ($P_{\text{local}} = 0.996$, $k = 3$) and *IDH2* ($P_{\text{local}} = 0.997$, $k = 4$) burst hubs, each independently radiating subclonal *FLT3*, *KRAS*, *NRAS*, and *PTPN11* signaling resistance branches. (**D**) Clonal trunk-to-subclone driver architecture across all 123 acute myeloid leukemia patients, separating early epigenetic/chromatin founders from late subclonal signaling resistance drivers (OR = 1.79×, two-sided Fisher $p = 1.9 \times 10^{-3}$; bubble size and numbers denote parallel within-patient acquisitions).

Grouping B-cell genotypes solely by how many mutant daughter lineages (k) radiate from each node predicts a 9.5-fold rise in in vivo clonal expansion from $k = 0$ terminal leaves (5.1%) to $k \geq 3$ burst hubs (48.2%; paired Wilcoxon $p = 5.9 \times 10^{-6}$), correlates with tighter yeast-display antigen binding within germinal centers ($p = 1.7 \times 10^{-3}$), and holds within exact mutational-depth strata ($p = 4.1 \times 10^{-6}$) while vanishing in neutral simulated repertoires ($p = 0.70$; Table S10). Across mice, mutations arising independently in five or more animals enrich 5.6-fold inside antigen-contacting complementarity-determining regions (64.0% versus 24.1%, $p =$

7.5×10^{-5}) and confer higher binding affinity ($p = 8.1 \times 10^{-5}$; Fig. 6A,B).

Extending the same framework beyond DNA sequences to somatic mutation and copy-number profiles in human cancer—with all cell counts and clinical labels withheld—reveals the same selective signature in patient tumors (Fig. 6A,C,D, Fig. S6, and Table S10). In a single leukemia patient (`AML-38-001`, 7,016 sequenced cells; Fig. 6C), reconstructing clonal history in 4.1 ms recovers an early *NPM1* founder that splits into parallel *IDH1* ($P_{\text{local}} = 0.996$, $k = 3$) and *IDH2* ($P_{\text{local}} = 0.997$, $k = 4$) burst hubs, each independently radiating subclonal *FLT3*, *KRAS*, *NRAS*, and *PTPN11* resistance branches, whereas Bayesian MCMC sampling (471 s) nests *IDH2* sequentially downstream of *IDH1* (Table S10). Across 168 COMPASS myeloid single-cell cohorts (129 patients, 1.00 million cells (*61*)), inferred burst hubs ($k \geq 3$) carry 6.18-fold more leukemic cells in vivo than terminal leaves (232.8 versus 37.7 cells; $p = 1.7 \times 10^{-18}$; exact patient-depth paired $p = 5.9 \times 10^{-12}$) and resolve 90.3% of stepwise copy-neutral loss-of-heterozygosity (CN-LOH) transitions; across the separate 123-patient Morita AML cohort (*32*), branching hubs spawn larger descendant lineages ($p = 3.0 \times 10^{-15}$) while late signaling resistance drivers enrich relative to early epigenetic founders (OR = 1.79, $p = 1.9 \times 10^{-3}$) and recur in parallel within 44 patients (Fig. 6A,D). On 4,950 single-cell whole-genome copy-number profiles across ovarian and breast tumors (*62*), ordered step coding compresses raw 500-kb genomes 21.9- to 33.5-fold in 22 s on colinear segments, contracting 39–53% of noisy splits while preserving cohort clade purity (57.9%–95.7% versus 20.8%–59.9% random null; `OV2295` ovarian tumor versus ascites 95.7% purity, compartment classification score $F_1 = 0.997$; Table S10).

Outlook

Viewing evolutionary inference as lossless compression shows that tree topology, branch support, evolutionary rates, gene transfer, and adaptive convergence follow from a single principle: shared descent shortens the description of biological variation. Because every branch or lateral link must pay its own cost in information, unresolved hubs become a quantitative biological signal, marking rapid species radiations, epidemic super-spreading, and Darwinian clonal bursts in immunity and cancer without arbitrary cutoffs. Pricing lateral links in the same currency separates local paralog gene conversion from cross-order convergence and turns debates over microbial gene transfer into genome-wide budgets, while enabling the inference of Tree-of-Life scale phylogenies on standard laptops and web browsers. Because this principle requires only heritable states copied with sparse changes, it extends naturally beyond sequences and somatic profiles to epigenetic marks, CRISPR lineage barcodes, and morphological or linguistic characters wherever history leaves a compressible trace.

References and Notes

1. Walter M. Fitch. Toward defining the course of evolution: minimum change for a specific tree topology. *Systematic Zoology*, 20(4):406–416, 1971. doi: 10.2307/2412116.

2. Joseph Felsenstein. Evolutionary trees from DNA sequences: a maximum likelihood approach. *Journal of Molecular Evolution*, 17(6):368–376, 1981. doi: 10.1007/BF01734359.

3. Naruya Saitou and Masatoshi Nei. The neighbor-joining method: a new method for reconstructing phylogenetic trees. *Molecular Biology and Evolution*, 4(4):406–425, 1987. doi: 10.1093/oxfordjournals.molbev.a040454.

4. Morgan N. Price, Paramvir S. Dehal, and Adam P. Arkin. FastTree 2 – approximately maximum-likelihood trees for large alignments. *PLoS ONE*, 5(3):e9490, 2010. doi: 10.1371/journal.pone.0009490.

5. Alexey M. Kozlov, Diego Darriba, Tomáš Flouri, Benoit Morel, and Alexandros Stamatakis. RAxML-NG: a fast, scalable and user-friendly tool for maximum likelihood phylogenetic inference. *Bioinformatics*, 35(21):4453–4455, 2019. doi: 10.1093/bioinformatics/btz305.

6. Bui Quang Minh, Heiko A. Schmidt, Olga Chernomor, Dominik Schrempf, Michael D. Woodhams, Arndt von Haeseler, and Robert Lanfear. IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. *Molecular Biology and Evolution*, 37(5):1530–1534, 2020. doi: 10.1093/molbev/msaa015.

7. Jorma Rissanen. Modeling by shortest data description. *Automatica*, 14(5):465–471, 1978. doi: 10.1016/0005-1098(78)90005-5.

8. Jorma Rissanen. A universal prior for integers and estimation by minimum description length. *The Annals of Statistics*, 11(2):416–431, 1983. doi: 10.1214/aos/1176346150.

9. Joseph Felsenstein. Confidence limits on phylogenies: an approach using the bootstrap. *Evolution*, 39(4):783–791, 1985. doi: 10.1111/j.1558-5646.1985.tb00420.x.

10. David Posada and Thomas R. Buckley. Model selection and model averaging in phylogenetics: advantages of Akaike information criterion and Bayesian approaches over likelihood ratio tests. *Systematic Biology*, 53(5):793–808, 2004. doi: 10.1080/10635150490522304.

11. Sergei L. Kosakovsky Pond, David Posada, Michael B. Gravenor, Christopher H. Woelk, and Simon D. W. Frost. Automated phylogenetic detection of recombination using a genetic algorithm. *Molecular Biology and Evolution*, 23(10):1891–1901, 2006. doi: 10.1093/molbev/msl051.

12. Antonis Rokas and Sean B. Carroll. Bushes in the tree of life. *PLoS Biology*, 4(11):e352, 2006. doi: 10.1371/journal.pbio.0040352.

13. Xing-Xing Shen, Chris Todd Hittinger, and Antonis Rokas. Contentious relationships in phylogenomic studies can be driven by a handful of genes. *Nature Ecology & Evolution*, 1 (5):0126, 2017. doi: 10.1038/s41559-017-0126.

14. Shiran Abadi, Dana Azouri, Tal Pupko, and Itay Mayrose. Model selection may not be a mandatory step for phylogeny reconstruction. *Nature Communications*, 10(1):934, 2019. doi: 10.1038/s41467-019-08822-w.

15. Joseph Felsenstein. Cases in which parsimony or compatibility methods will be positively misleading. *Systematic Zoology*, 27(4):401–410, 1978. doi: 10.2307/2412923.

16. Simon Tavaré. Some probabilistic and statistical problems in the analysis of DNA sequences. In *Lectures on Mathematics in the Life Sciences*, volume 17, pages 57–86. American Mathematical Society, Providence, RI, 1986.

17. Ziheng Yang. Maximum likelihood phylogenetic estimation from DNA sequences with variable rates over sites: approximate methods. *Journal of Molecular Evolution*, 39(3):306–314, 1994. doi: 10.1007/BF00160154.

18. Si Quang Le and Olivier Gascuel. An improved general amino acid replacement matrix. *Molecular Biology and Evolution*, 25(7):1307–1320, 2008. doi: 10.1093/molbev/msn067.

19. W. Ford Doolittle. Phylogenetic classification and the universal tree. *Science*, 284(5423): 2124–2128, 1999. doi: 10.1126/science.284.5423.2124.

20. Michael Worobey, Andrew Rambaut, and Edward C. Holmes. Widespread intra-serotype recombination in natural populations of dengue virus. *Proceedings of the National Academy of Sciences*, 96(13):7352–7357, 1999. doi: 10.1073/pnas.96.13.7352.

21. Peter Parham and Tomoko Ohta. Population biology of antigen presentation by MHC class I molecules. *Science*, 272(5258):67–74, 1996. doi: 10.1126/science.272.5258.67.

22. Yang Liu, James A. Cotton, Bin Shen, Xiuqun Han, Stephen J. Rossiter, and Shuyi Zhang. Convergent sequence evolution between echolocating bats and dolphins. *Current Biology*, 20(2):R53–R54, 2010. doi: 10.1016/j.cub.2009.11.058.

23. Ying Li, Zhen Liu, Peng Shi, and Jianzhi Zhang. The hearing gene Prestin unites echolocating bats and whales. *Current Biology*, 20(2):R55–R56, 2010. doi: 10.1016/j.cub.2009.11.042.

24. Xiaofan Zhou, Xing-Xing Shen, Chris Todd Hittinger, and Antonis Rokas. Evaluating fast maximum likelihood-based phylogenetic programs using empirical phylogenomic data sets. *Molecular Biology and Evolution*, 35(2):486–503, 2018. doi: 10.1093/molbev/msx302.

25. Joe Parker, Georgia Tsagkogeorga, James A. Cotton, Yuan Liu, Paolo Provero, Elia Stupka, and Stephen J. Rossiter. Genome-wide signatures of convergent evolution in echolocating mammals. *Nature*, 502(7470):228–231, 2013. doi: 10.1038/nature12511.

26. Zhengting Zou and Jianzhi Zhang. No genome-wide protein sequence convergence for echolocation. *Molecular Biology and Evolution*, 32(5):1237–1241, 2015. doi: 10.1093/molbev/msv014.

27. Yatish Turakhia, Bryan Thornlow, Angie Hinrichs, Jakob McBroome, Nicolas Ayala, Cheng Ye, Kyle Smith, Nicola De Maio, David Haussler, Robert Lanfear, and Russell Corbett-Detig. Pandemic-scale phylogenomics reveals the SARS-CoV-2 recombination landscape. *Nature*, 609(7929):994–997, 2022. doi: 10.1038/s41586-022-05189-9.

28. Yury V. Malovichko, Bernhard Bein, Alejandro Gonzales-Irribarren, Evgeny Leushkin, Leon Hilgers, Amy Stephen, Xueling Yi, Michele Albertini, Tim Stadager, Markus Zumpt, Luca Hoppach, Felix Götz, Niklas Himstedt, Lucas Koch, Vertebrate Genomes Project Consortium Phase I, and Michael Hiller. Accurate, comprehensive gene annotation and ortholog identification across thousands of vertebrate genomes with TOGA2. *bioRxiv*, 2026. doi: 10.64898/2026.06.30.735536.

29. Ravi Jain, Maria C. Rivera, and James A. Lake. Horizontal gene transfer among genomes: the complexity hypothesis. *Proceedings of the National Academy of Sciences*, 96(7):3801–3806, 1999. doi: 10.1073/pnas.96.7.3801.

30. Donovan H. Parks, Maria Chuvochina, Pierre-Alain Chaumeil, Christian Rinke, Aaron J. Mussig, and Philip Hugenholtz. A complete domain-to-species taxonomy for Bacteria and Archaea. *Nature Biotechnology*, 38(9):1079–1086, 2020. doi: 10.1038/s41587-020-0501-8.

31. William S. DeWitt, Ashni A. Vora, Tatsuya Araki, Jared G. Galloway, Tanwee Alkutkar, Juliana Bortolatto, Tiago B. R. Castro, Will Dumm, Chris Jennings-Shaffer, Tongqiu Jia, Luka Mesin, Jesse D. Bloom, Gabriel Ozorowski, Juhee Pae, Duncan K. Ralph, Armita Nourmohammad, Yun S. Song, Andrew B. Ward, Tyler N. Starr, Frederick A. Matsen, IV, and Gabriel D. Victora. Replaying germinal center evolution on a quantified affinity landscape. *Cell*, 2026.

32. Kiyomi Morita, Feng Wang, Katharina Jahn, Tianyuan Hu, Tomoyuki Tanaka, Yuya Sasaki, Jack Kuipers, Sanam Loghavi, Sa A. Wang, Yan Yan, Ken Furudate, Jairo Matthews, Latasha Little, Curtis Gumbs, Jianhua Zhang, Xingzhi Song, Erika Thompson, Keyur P. Patel, Carlos E. Bueso-Ramos, Courtney D. DiNardo, Farhad Ravandi, Elias Jabbour, Michael Andreeff, Jorge Cortes, Kapil Bhalla, Guillermo Garcia-Manero, Hagop Kantarjian, Marina Konopleva, Daisuke Nakada, Nicholas Navin, Niko Beerenwinkel, P. Andrew

Futreal, and Koichi Takahashi. Clonal evolution of acute myeloid leukemia revealed by high-throughput single-cell genomics. *Nature Communications*, 11(1):5327, 2020. doi: 10.1038/s41467-020-19119-8.

33. Margaret O. Dayhoff, Robert M. Schwartz, and Bruce C. Orcutt. A model of evolutionary change in proteins. In Margaret O. Dayhoff, editor, *Atlas of Protein Sequence and Structure*, volume 5 of *Supplement 3*, pages 345–352. National Biomedical Research Foundation, Washington, DC, 1978.

34. Nhan Ly-Trong, Suha Naser-Khdour, Robert Lanfear, and Bui Quang Minh. AliSim: a fast and versatile phylogenetic sequence simulator for the genomic era. *Molecular Biology and Evolution*, 39(5):msac092, 2022. doi: 10.1093/molbev/msac092.

35. Michael Gerth. Neglecting model selection alters phylogenetic inference. *bioRxiv*, page 849018, 2019. doi: 10.1101/849018.

36. James H. Degnan and Noah A. Rosenberg. Discordance of species trees with their most likely gene trees. *PLoS Genetics*, 2(5):e68, 2006. doi: 10.1371/journal.pgen.0020068.

37. Chao Zhang, Rasmus Nielsen, and Siavash Mirarab. ASTER: A package for large-scale phylogenomic reconstructions. *Molecular Biology and Evolution*, 42(8):msaf172, 2025. doi: 10.1093/molbev/msaf172.

38. Erich D. Jarvis, Siavash Mirarab, Andre J. Aberer, Bo Li, Peter Houde, Cai Li, Simon Y. W. Ho, Brant C. Faircloth, Benoit Nabholz, Jason T. Howard, et al. Whole-genome analyses resolve early branches in the tree of life of modern birds. *Science*, 346(6215):1320–1331, 2014. doi: 10.1126/science.1253451.

39. Richard O. Prum, Jacob S. Berv, Alex Dornburg, Daniel J. Field, Jeffrey P. Townsend, Emily Moriarty Lemmon, and Alan R. Lemmon. A comprehensive phylogeny of birds (Aves) using targeted next-generation DNA sequencing. *Nature*, 526(7574):569–573, 2015. doi: 10.1038/nature15697.

40. Norman J. Wickett, Siavash Mirarab, Nam Nguyen, Tandy Warnow, Eric Carpenter, Naim Matasci, Saravanaraj Ayyampalayam, Michael S. Barker, J. Gordon Burleigh, Matthew A. Gitzendanner, et al. Phylotranscriptomic analysis of the origin and early diversification of land plants. *Proceedings of the National Academy of Sciences*, 111(45):E4859–E4868, 2014. doi: 10.1073/pnas.1323926111.

41. Bernhard Misof, Shanlin Liu, Karen Meusemann, Ralph S. Peters, Alexander Donath, Christoph Mayer, Paul B. Frandsen, Jessica Ware, Tomáš Flouri, Rolf G. Beutel, et al. Phylogenomics resolves the timing and pattern of insect evolution. *Science*, 346(6210): 763–767, 2014. doi: 10.1126/science.1257570.

42. Meng-Yun Chen, Dan Liang, and Peng Zhang. Selecting question-specific genes to reduce incongruence in phylogenomics: a case study of jawed vertebrate backbone phylogeny. *Systematic Biology*, 64(6):1104–1120, 2015. doi: 10.1093/sysbio/syv059.

43. Marek L. Borowiec, Ernest K. Lee, Joanna C. Chiu, and David C. Plachetzki. Extracting phylogenetic signal and accounting for bias in whole-genome data sets supports the Ctenophora as sister to remaining Metazoa. *BMC Genomics*, 16:987, 2015. doi: 10.1186/s12864-015-2146-4.

44. Nathan V. Whelan, Kevin M. Kocot, Tatiana P. Moroz, Krishanu Mukherjee, Peter Williams, Gustav Paulay, Leonid L. Moroz, and Kenneth M. Halanych. Ctenophore relationships and their placement as the sister group to all other animals. *Nature Ecology & Evolution*, 1(11):1737–1746, 2017. doi: 10.1038/s41559-017-0331-3.

45. Sen Song, Liang Liu, Scott V. Edwards, and Shaoyuan Wu. Resolving conflict in eutherian mammal phylogeny using phylogenomics and the multispecies coalescent model. *Proceedings of the National Academy of Sciences*, 109(37):14942–14947, 2012. doi: 10.1073/pnas.1211733109.

46. Allison J. Shultz and Timothy B. Sackton. Immune genes are hotspots of shared positive selection across birds and mammals. *eLife*, 8:e41815, 2019. doi: 10.7554/eLife.41815.

47. Bogdan M. Kirilenko, Chetan Munegowda, Ekaterina Osipova, David Jebb, Virag Sharma, Moritz Blumer, Ariadna E. Morales, AW Ahmed, Dimitrios-Georgios Kontopoulos, Leon Hilgers, Kerstin Lindblad-Toh, Elinor K. Karlsson, Zoonomia Consortium, and Michael Hiller. Integrating gene annotation with orthology inference at scale. *Science*, 380(6643):eabn3107, 2023. doi: 10.1126/science.abn3107.

48. Gregg W. C. Thomas and Matthew W. Hahn. Determining the null model for detecting adaptive convergence from genomic data: a case study using echolocating mammals. *Molecular Biology and Evolution*, 32(5):1232–1236, 2015. doi: 10.1093/molbev/msv013.

49. Pascal-Antoine Christin, Erika J. Edwards, Guillaume Besnard, Susanna F. Boxall, Richard Gregory, Elizabeth A. Kellogg, James Hartwell, and Colin P. Osborne. Adaptive evolution of C_4 photosynthesis through recurrent lateral gene transfer. *Current Biology*, 22(5):445–449, 2012. doi: 10.1016/j.cub.2012.01.054.

50. Jean K. Carr, Mika O. Salminen, Jan Albert, Eric Sanders-Buell, D. Gotte, Deborah L. Birx, and Francine E. McCutchan. Full genome sequences of human immunodeficiency virus type 1 subtypes G and A/G intersubtype recombinants. *Virology*, 247(1):22–31, 1998. doi: 10.1006/viro.1998.9211.

51. Ian H. Witten, Radford M. Neal, and John G. Cleary. Arithmetic coding for data compression. *Communications of the ACM*, 30(6):520–540, 1987. doi: 10.1145/214762.214771.

52. Maciej F. Boni, Philippe Lemey, Xiaowei Jiang, Tommy Tsan-Yuk Lam, Blair W. Perry, Todd A. Castoe, Andrew Rambaut, and David L. Robertson. Evolutionary origins of the SARS-CoV-2 sarbecovirus lineage responsible for the COVID-19 pandemic. *Nature Microbiology*, 5(11):1408–1417, 2020. doi: 10.1038/s41564-020-0771-4.

53. Spyros Lytras, Joseph Hughes, Darren Martin, Phillip Swanepoel, Alice de Klerk, Rentia Lourens, Sergei L. Kosakovsky Pond, Wei Xia, Xiaowei Jiang, and David L. Robertson. Exploring the natural origins of SARS-CoV-2 in the light of recombination. *Genome Biology and Evolution*, 14(2):evac018, 2022. doi: 10.1093/gbe/evac018.

54. Donovan H. Parks, Maria Chuvochina, David W. Waite, Christian Rinke, Adam Skarshewski, Pierre-Alain Chaumeil, and Philip Hugenholtz. A standardized bacterial taxonomy based on genome phylogeny substantially revises the tree of life. *Nature Biotechnology*, 36(10):996–1004, 2018. doi: 10.1038/nbt.4229.

55. Carl R. Woese, Gary J. Olsen, Michael Ibba, and Dieter Söll. Aminoacyl-tRNA synthetases, the genetic code, and the evolutionary process. *Microbiology and Molecular Biology Reviews*, 64(1):202–236, 2000. doi: 10.1128/MMBR.64.1.202-236.2000.

56. Kelly L. Wyres, Margaret M. C. Lam, and Kathryn E. Holt. Population genomics of *Klebsiella pneumoniae*. *Nature Reviews Microbiology*, 18(6):344–359, 2020. doi: 10.1038/s41579-019-0315-1.

57. Cindy J. Castelle and Jillian F. Banfield. Major new microbial groups expand diversity and alter our understanding of the tree of life. *Cell*, 172(6):1181–1197, 2018. doi: 10.1016/j.cell.2018.02.016.

58. Daniel McDonald, Morgan N. Price, Julia Goodrich, Eric P. Nawrocki, Todd Z. DeSantis, Alexander Probst, Gary L. Andersen, Rob Knight, and Philip Hugenholtz. An improved Greengenes taxonomy with explicit ranks for ecological and evolutionary analyses of bacteria and archaea. *The ISME Journal*, 6(3):610–618, 2012. doi: 10.1038/ismej.2011.139.

59. Jeroen M. J. Tas, Luka Mesin, Giulia Pasqual, Sasha Targ, Johanne T. Jacobsen, Yasuko M. Mano, Casie S. Chen, Jean-Claude Weill, Claude-Agnès Reynaud, Edward P. Browne, Michael Meyer-Hermann, and Gabriel D. Victora. Visualizing antibody affinity maturation in germinal centers. *Science*, 351(6277):1048–1054, 2016. doi: 10.1126/science.aad3439.

60. William S. DeWitt, Luka Mesin, Gabriel D. Victora, Vladimir N. Minin, and Frederick A. Matsen, IV. Using genotype abundance to improve phylogenetic inference. *Molecular Biology and Evolution*, 35(5):1253–1265, 2018. doi: 10.1093/molbev/msy020.

61. Etienne Sollier, Jack Kuipers, Koichi Takahashi, Niko Beerenwinkel, and Katharina Jahn. COMPASS: joint copy number and mutation phylogeny reconstruction from amplicon single-cell sequencing data. *Nature Communications*, 14(1):4921, 2023. doi: 10.1038/s41467-023-40378-8.

62. Sohrab Salehi, Fatemeh Dorri, Kevin Chern, Farhia Kabeer, Nicole Rusk, Tyler Funnell, Marc J. Williams, Daniel Lai, Mirela Andronescu, Kieran R. Campbell, Andrew McPherson, Samuel Aparicio, Andrew Roth, Sohrab P. Shah, and Alexandre Bouchard-Côté. Cancer phylogenetic tree inference at scale from 1000s of single cell genomes. *Peer Community Journal*, 3:e63, 2023. doi: 10.24072/pcjournal.292.

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Competing interests: The authors declare that they have no competing interests.
Data and materials availability: The open-source Rust and WebAssembly implementation of Bushwhacker (supporting nucleotide, amino-acid, binary, and discrete/ordered character matrices), the `.bwz` lossless archival codec, interactive browser application (`https://veg.github.io/bushwacker/`), reproducibility scripts, and empirical and ground-truth validation ledgers spanning more than 81,000 empirical alignments, 1.01 million single cells, and 37,020 simulated evolutionary histories are available at `https://github.com/veg/bushwacker` and bundled in the Zenodo v1.0 replication archive (`bushwhacker_science_zenodo_v1.0`, containing source code, specimens, assembled alignments, all JSON/CSV benchmark ledgers, figure scripts, and automated downloaders for external public archives; see Materials and Methods for complete accessions and DOIs).

Supplementary Materials

Materials and Methods
Supplementary Text
Figs. S1 to S6
Tables S1 to S11
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